

# Introduction to SMART Wi-Fi

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We need faster tools and media to prove our ideas, and this is where ready-to-use modules like Smart Wi-Fi enter. Fig. 1 shows the top side of Smart Wi-Fi module.

The module is low-cost and versatile; all a developer needs is a USB cable and it is ready to work. For



Fig. 1: Smart Wi-Fi module

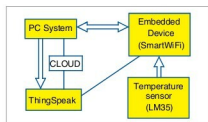


Fig. 2 Block diagram of the introduction of Smart Wi-Fi

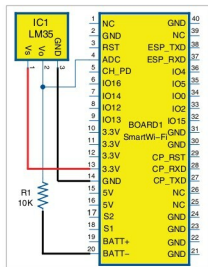


Fig. 3: Circuit diagram of the Smart Wi-Fi

software developers, working with hardware is always a pain, and this is one of the biggest reasons why many ideas never exit a computer. Smart Wi-Fi enables making a product quickly and reliably. With open software resources and hardware data, moving to the final product after the proof of concept is also easy.

One of the major aims for writing this article is to show the versatility of this product and show how quick the development can be. In this article a few tests with Smart Wi-Fi are discussed, which show the applicability of this module for development of the Internet of Things (IoT) products.

## Circuit and working

The circuit diagram of the Smart Wi-Fi is shown in Fig. 3. It is built around Smart Wi-Fi module

### PARTS LIST

|                                                          |                   |
|----------------------------------------------------------|-------------------|
| <b>Semiconductors:</b>                                   |                   |
| BOARD1                                                   | - Smart Wi-Fi     |
| IC1                                                      | - LM35            |
| <b>Resistor (1/4-watt, <math>\pm 5\%</math> carbon):</b> |                   |
| R1                                                       | - 10-kilo-ohm     |
| <b>Miscellaneous:</b>                                    |                   |
|                                                          | - Micro-USB cable |

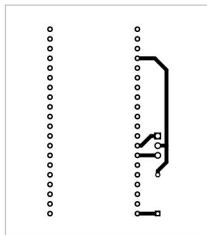


Fig. 4: Actual-size PCB pattern of Smart Wi-Fi

(BOARD1), micro-USB cable, LM35 temperature sensor (IC1) and a resistor (R1).

Smart Wi-Fi has ESP8266 Wi-Fi module along with an onboard voltage regulator, USB port, battery charger, user button and debug RGB LED.

Temperature is sensed by IC1 and its output is measured with ADC pin of BOARD1. BOARD1 sends data to the serial port. Data can be seen in the serial terminal of ESPlorer tool.

Real-time temperature data is uploaded over ThingSpeak ([www.thingspeak.com](http://www.thingspeak.com)).

## Construction and testing

An actual-size, single-side PCB of Smart Wi-Fi is shown in Fig. 4 and its component layout in Fig. 5.

ESP8266 module has many tools that are freely available, and you can select these as per your requirement. For a person new to this kind of development, following tools are recommended. Download and install these as per your PC configuration.

1. ESPlorer for serial port interface with Smart Wi-Fi (<http://esp8266.ru/esplorer/>)
2. CP210x USB drivers ([www.silabs.com](http://www.silabs.com)).

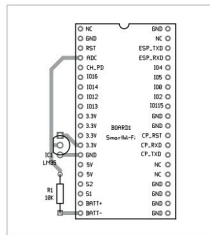


Fig. 5: Component layout of the PCB